Activity

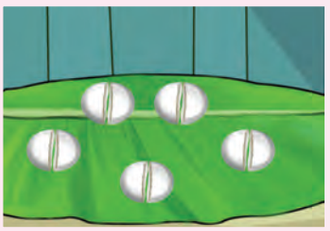
Using the given fractions $\frac{1}{5}, \frac{1}{6}, \frac{1}{10}, \frac{1}{15}, \frac{2}{15}, \frac{4}{15}, \frac{1}{30}, \frac{7}{30}$ and $\frac{9}{30}$ fill in the missing ones in the given 3×3 square in such a way that the addition of fractions through rows, columns and diagonals give the same total $\frac{1}{2}$.

	$\frac{1}{30}$	
	$\frac{1}{6}$	
$\frac{2}{15}$		

1.4 Improper and Mixed Fractions

Think about the situation





Iniyan had 5 idlis for his breakfast. When he was about to eat, his friend Abdul came. He wanted to share it equally with his friend Abdul. Both of them have taken 2 each and $\frac{1}{2}$ of the remaining idli.

Each one has eaten 2 full idlis and $\frac{1}{2}$ idli. This can be represented as $2+\frac{1}{2}=2\frac{1}{2}$. This representation is called a ***mixed fraction***. Thus, a ***mixed fraction*** is the sum of a whole number and a proper fraction. Also we can express an improper fraction as a mixed fraction by dividing the numerator by denominator to get quotient and remainder. Thus, any mixed fraction can be written as

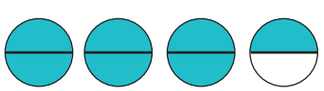
$$\text{Quotient} + \frac{\text{Remainder}}{\text{Divisor}} = \text{Quotient}\,\frac{\text{Remainder}}{\text{Divisor}}.$$

Another way to share these idlis is as follows: Now can you see how many halves are there in 5 idlis. There are 10 halves. If we share these $\frac{1}{2}$ idlis each time, then Iniyan and Abdul has eaten 5 halves each. That is $\frac{1}{2}+\frac{1}{2}+\frac{1}{2}+\frac{1}{2}+\frac{1}{2}=\frac{5}{2}$ which is same as $2\frac{1}{2}=2+\frac{1}{2}=\frac{(2\times 2)+1}{2}=\frac{5}{2}$. Thus, any improper fraction can be written as mixed fraction as

$$\text{Improper fraction} = \frac{\text{(Whole number X Denominator) + Numberator}}{\text{Denominator}}.$$

Try these

1. Complete the following table. The first one is done for you.

Mixed Fraction	Diagrams	Improper Fraction
i) **3** circles are **completely shaded.** $\frac{1}{2}$ of another circle is shaded. That is totally $3\frac{1}{2}$ circles are shaded.		Each circle is divided into halves. There are totally **7 half circles shaded** which is equal to $\frac{7}{2}$.

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